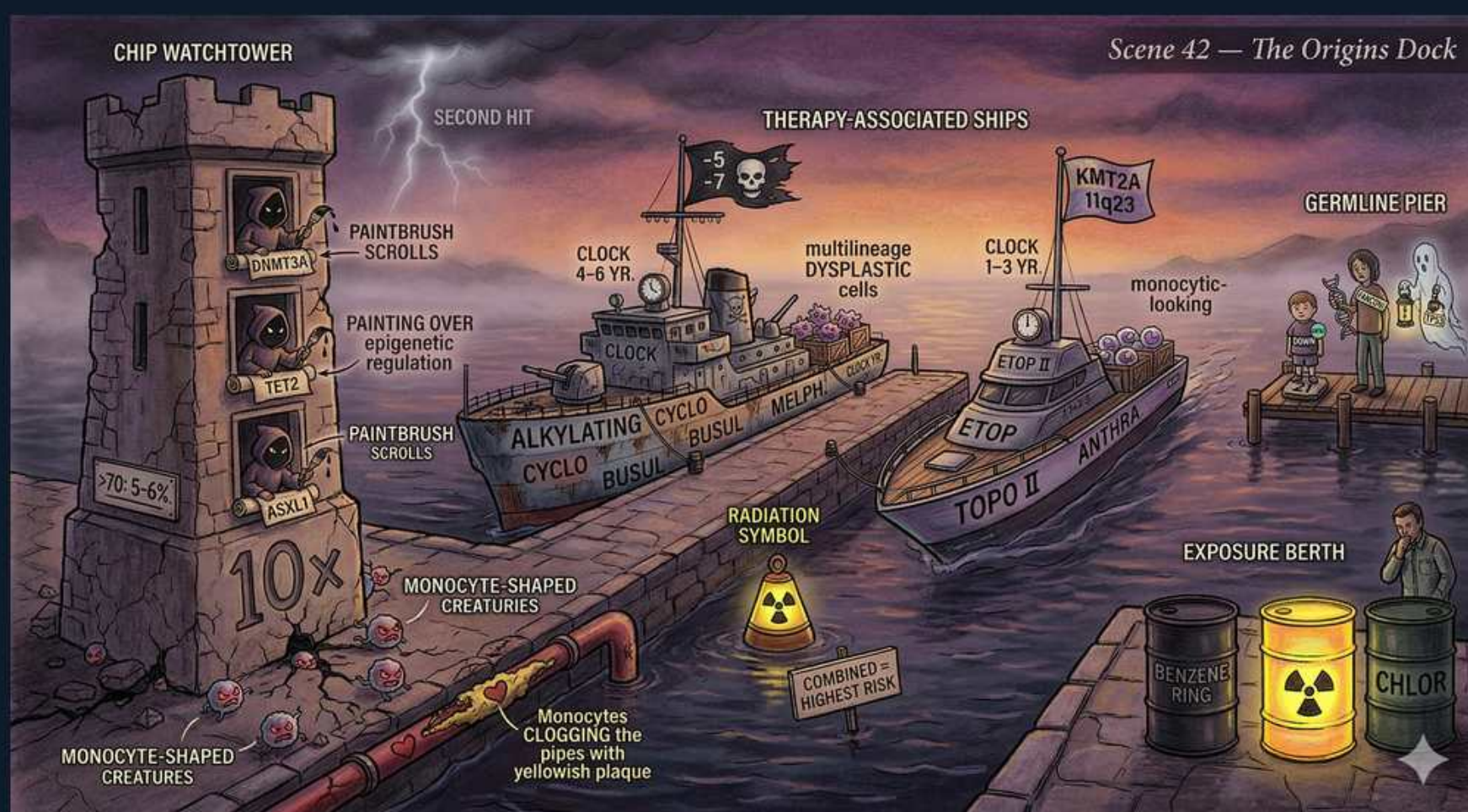


AML — Secondary AML & Etiology



1. Therapy-related ships

WHAT YOU SEE

A banner reading THERAPY-ASSOCIATED SHIPS over two warships at the dock — the fleet is therapy-related AML, uniformly adverse risk.

WHAT IT ENCODES

Therapy-related AML (t-AML) arises years after cytotoxic chemotherapy or radiation given for a prior malignancy or autoimmune disease and is uniformly classified as ADVERSE risk with poor response to standard therapy. The two archetypes are alkylator-induced (long latency, -5/-7, complex karyotype, antecedent MDS phase) and topoisomerase-II-induced (short latency, KMT2A/11q23, no MDS phase). CPX-351 is the preferred induction in older t-AML patients.

BOARD TRAP

All therapy-related AML is adverse-risk regardless of cytogenetics and is best considered for allo-HCT/clinical trial; CPX-351 outperforms conventional 7+3 here. Distinguishing the two t-AML types by latency and genetics is high-yield.

2. Alkylating ship 4-6 yr

WHAT YOU SEE

A warship hulled ALKYLATING (CYCLO/BUSUL/MELPH) flying a clock reading 4-6 YR — the slow-clock ship is alkylator t-AML with long latency and -5/-7.

WHAT IT ENCODES

Alkylating agents (cyclophosphamide, busulfan, melphalan, chlorambucil, platinum) cause t-AML after a LONG latency of about 5-7 years, typically preceded by a myelodysplastic (MDS) phase, and carry unbalanced deletions — monosomy 5/del(5q), monosomy 7/del(7q) — and complex karyotypes, often with TP53 mutation. This is the worse-prognosis of the two t-AML patterns.

BOARD TRAP

Long latency (~5-7 yr) + preceding MDS + del(5q)/del(7q)/complex karyotype = ALKYLATOR-induced t-AML. Pairing the wrong agent class with these features is the classic distractor.

3. Topo II ship 1-3 yr

WHAT YOU SEE

A warship hulled TOPO II (ETOP/ANTHRA) flying a clock reading 1-3 YR — the fast-clock ship is topo-II t-AML with short latency and KMT2A.

WHAT IT ENCODES

Topoisomerase II inhibitors (etoposide, teniposide, anthracyclines) cause t-AML after a SHORT latency of about 1-3 years, with NO preceding MDS phase, and characteristically harbor balanced translocations involving KMT2A/MLL at 11q23 (or sometimes RUNX1/CBF rearrangements). This monocytic-leaning leukemia appears more abruptly than the alkylator type.

BOARD TRAP

Short latency (~1-3 yr), no MDS phase, balanced KMT2A/11q23 translocation = TOPO-II-induced t-AML — the mirror image of the alkylator pattern. The latency period is the quickest discriminator on the exam.

4. KMT2A 11q23

WHAT YOU SEE

A flag reading KMT2A 11q23 flying on the Topo II ship — the mast flag is the 11q23 rearrangement signature of topo-II t-AML.

WHAT IT ENCODES

KMT2A (MLL) rearrangement at chromosome band 11q23 is the molecular signature of topoisomerase-II-inhibitor-related AML and is itself an adverse-risk lesion (except the intermediate t(9;11)). KMT2A rearrangements also occur in de novo infant leukemias and confer chemoresistance.

BOARD TRAP

11q23/KMT2A links specifically to TOPO-II t-AML and is adverse — except t(9;11)(p21;q23)/KMT2A-MLLT3, the single intermediate-risk KMT2A rearrangement.

5. Multilineage dysplasia

WHAT YOU SEE

Misshapen multilineage dysplastic cells heaped on the alkylating ship's deck — the deformed cells are MDS-related changes marking secondary AML.

WHAT IT ENCODES

AML with myelodysplasia-related changes — defined by an antecedent MDS/MDS-MPN, MDS-related cytogenetics, or multilineage dysplasia, often with secondary-type mutations (SRSF2, SF3B1, U2AF1, ZRSR2, ASXL1, EZH2, BCOR, STAG2) — is a secondary-type AML that is adverse-risk and responds poorly to conventional chemo. Like t-AML, older patients benefit from CPX-351.

BOARD TRAP

AML with myelodysplasia-related changes / secondary-type mutations is adverse-risk secondary AML, biologically distinct from de novo AML, and warrants transplant-directed or HMA-based strategies rather than expecting a good 7+3 response.

6. Epigenetic mutations

WHAT YOU SEE

A watchtower with DNMT3A, TET2, ASXL1 scrolls and shadowy figures (CHIP WATCHTOWER) — the tower of 'DTA' scrolls is clonal hematopoiesis, the premalignant founding lesions.

WHAT IT ENCODES

DNMT3A, TET2, and ASXL1 ('DTA') mutations are epigenetic regulators and the founding lesions of clonal hematopoiesis of indeterminate potential (CHIP) — clonal expansion without overt malignancy that raises the long-term risk of AML/MDS and of cardiovascular disease. ASXL1 in particular is an adverse-risk marker, and DTA mutations persist in remission so they are unreliable MRD markers.

BOARD TRAP

DTA mutations (DNMT3A/TET2/ASXL1) persist after remission and therefore are POOR MRD markers — their presence in CR does not equal residual disease. CHIP is a premalignant state, not leukemia, but predicts transformation and cardiovascular risk.

7. Monocytic clogging

WHAT YOU SEE

Round monocyte-shaped creatures clogging the pier pipes with yellow plaque — the gunk-clogging monocytes are M4/M5 monocytic AML infiltrating tissues.

WHAT IT ENCODES

Monocytic differentiation (FAB M4 myelomonocytic and M5 monoblastic/monocytic) drives tissue infiltration — gingival hypertrophy, leukemia cutis, CNS involvement — and a higher leukostasis risk. Monocytic blasts express CD14/CD11b/CD64 and are nonspecific-esterase positive (versus MPO-strong granulocytic blasts).

BOARD TRAP

Monocytic AML (M4/M5) is the subtype behind gum infiltration, skin involvement, and elevated leukostasis/CNS risk — and is distinguished by nonspecific esterase positivity rather than strong MPO.

8. Germline pier

WHAT YOU SEE

A pier signed GERMLINE with a family and a ghost figure — the ancestral pier is inherited predisposition (RUNX1/CEBPA/GATA2/DDX41) in relatives.

WHAT IT ENCODES

Germline predisposition syndromes (RUNX1, CEBPA, GATA2, DDX41, and bone marrow failure syndromes such as Fanconi anemia, dyskeratosis congenita, telomere disorders) cause familial MDS/AML. Recognizing them matters because affected relatives must be excluded as stem-cell donors and because surveillance is needed — testing requires germline (e.g., cultured fibroblast/skin) DNA, not blood, since blood is clonally involved.

BOARD TRAP

Before using a RELATIVE as a transplant donor, screen for germline predisposition (RUNX1/CEBPA/GATA2/DDX41) — transplanting a sibling's predisposed marrow can transmit/recur leukemia. Germline testing needs non-hematopoietic tissue (fibroblasts), not a blood sample.

9. Benzene barrel

WHAT YOU SEE

A hazard barrel stamped BENZENE (benzene-ring icon) on the exposure berth — the toxic drum is benzene, the best-established chemical AML cause.

WHAT IT ENCODES

Benzene is the best-established occupational/environmental chemical cause of AML (and MDS/aplastic anemia), with risk related to cumulative exposure in industries like petroleum, rubber, and tanning, and from cigarette smoke. It causes clastogenic marrow damage leading to secondary-type cytogenetic abnormalities.

BOARD TRAP

Benzene (and chronic cigarette smoking) is a confirmed AML risk factor — an occupational history in petroleum/rubber/solvent work is the testable clue. Toluene and similar solvents are not equivalently linked.

10. Radiation symbol

WHAT YOU SEE

A glowing yellow radiation-hazard buoy reading COMBINED = HIGHEST RISK — the radioactive buoy is ionizing radiation, worst when combined with alkylators.

WHAT IT ENCODES

Ionizing radiation (atomic-bomb survivors, radiotherapy, prior radioiodine) is an established AML cause; the combination of radiation plus alkylating chemotherapy yields the highest secondary-AML risk of all. Latency is typically several years, and the resulting leukemia carries adverse-risk genetics.

BOARD TRAP

Combined chemoradiation carries the highest t-AML risk — greater than either modality alone. CLL/lymphoma survivors treated with chemo + radiation are a classic high-risk population for later AML.

11. Chlor / chemical exposure

WHAT YOU SEE

A hazard barrel stamped CHLOR at the exposure berth — the chemical drum is additional occupational/cytotoxic exposures contributing to secondary AML.

WHAT IT ENCODES

Beyond benzene, additional chemical/cytotoxic exposures contribute to secondary AML, and patients with a prior cancer treated with cytotoxic agents remain at lifelong elevated risk. A careful occupational, environmental, and prior-treatment history is part of every new AML workup to identify a secondary etiology.

BOARD TRAP

A secondary etiology (chemical, occupational, prior chemo/radiation) reclassifies AML as adverse-risk and changes management — always take the exposure and prior-treatment history, since de novo and secondary AML are treated differently.

12. 10x risk multiplier

WHAT YOU SEE

A 10x marker carved into the watchtower base — the tenfold multiplier is an antecedent MDS/MPN markedly raising AML risk (secondary AML).

WHAT IT ENCODES

An antecedent hematologic disorder — MDS, MPN (e.g., myelofibrosis, PV, ET), aplastic anemia, or PNH — markedly increases the risk of progression to AML and defines secondary AML, which is adverse-risk with worse outcomes than de novo disease. MPN-blast-phase ($\geq 20\%$ blasts) AML is especially treatment-resistant.

BOARD TRAP

AML evolving from MDS/MPN ('secondary AML') is adverse-risk and responds poorly to standard 7+3 — these patients are steered toward HMA-based therapy, CPX-351, or transplant rather than expecting de-novo-level cure rates.